

1. (a) Basis for $\text{Col}(A) = \left\{ \begin{bmatrix} 1 \\ 3 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 5 \\ 2 \\ 8 \end{bmatrix} \right\}$

(b) Basis for $\text{Nul}(A) = \left\{ \begin{bmatrix} -4 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

(c) $\begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ is not in $\text{Col}(A)$. Augment A by $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and row reduce. We see there is a pivot in the augmented column so the system $A\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ is inconsistent.

2. Write $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ as a linear comb. of the basis vectors. Weights = entries in the coordinate vectors.

a. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{B_1} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{B_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, (c) $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{B_3} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$

3. A is $m \times n$ (a) True (b) True (must be sums of squares)

(c) True (must be a free variable) (d) False $\det A$ is not defined if A is not square.

4. $\begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$

5. $\text{Nul}(A) = \{\vec{u} \in \mathbb{R}^n : A\vec{u} = \vec{0}\}$ is a subset of $\mathbb{R}^n$ since A has n columns.

$\vec{0} \in \text{Nul}(A)$ because $A\vec{0} = \vec{0}$. $\therefore \text{Nul}(A)$ is not empty

Let $u, v \in \text{Nul}(A)$. Then $A\vec{u} = \vec{0}$ and $A\vec{v} = \vec{0}$

$\therefore A(u+v) = A\vec{u} + A\vec{v} = \vec{0} + \vec{0} = \vec{0} \quad \therefore u+v \in \text{Nul}(A)$

so $\text{Nul}(A)$ is closed under addition.

Let c be any scalar and $u \in \text{Nul}(A)$

$A \cdot c\vec{u} = c \cdot A\vec{u} = c \cdot \vec{0} = \vec{0} \quad \therefore c\vec{u} \in \text{Nul}(A)$

so $\text{Nul}(A)$ is closed under scalar multiplication

$\therefore \text{Nul}(A)$ is a subspace.

6. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$.

$$\text{Trace}(A+B) = \text{Trace}\left(\begin{bmatrix} a+e & b+f \\ c+g & d+h \end{bmatrix}\right) = a+e+d+h = (a+d) + e+h \\ = \text{Trace}(A) + \text{Trace}(B)$$

∴ Trace preserves addition.

Let k be any scalar.

$$\text{Trace}(kA) = \text{Trace}\left(\begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}\right) = ka+kd = k(a+d) = k \cdot \text{Trace}(A)$$

∴ Trace preserves scalar multiplication.

∴ Trace is a linear transformation.

7. determinant $= 0$ when $\lambda = 1, 4$, or -6 . since this determinant is just the product of the diagonal elements.

$$8. \begin{vmatrix} 3 & 4 & 2 & -2 \\ 0 & 5 & 0 & -4 \\ 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & -1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 4 & 2 & -2 \\ 5 & 0 & -4 \\ 2 & 0 & -1 \end{vmatrix} = 1 \cdot (-2) \cdot \begin{vmatrix} 5 & -4 \\ 2 & -1 \end{vmatrix} = -2(-5+8) = \boxed{-6}$$

9. A is 4×3 and B is 3×2 so AB is 4×2 .

$$(b) \text{ 1st Column of } AB = 1 \begin{bmatrix} 1 \\ 3 \\ -1 \\ 0 \end{bmatrix} + 0 \begin{bmatrix} 0 \\ 2 \\ -2 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 1 \\ 10 \end{bmatrix}$$

$$(c) \text{ 2nd Row of } AB = 3 \cdot [1, 2] + 2[0, 3] + 1[2, -1] = [5, 11]$$

$$(d) AB = \text{Col } 1 \cdot \text{Row } 1 + \text{Col } 2 \cdot \text{Row } 2 + \text{Col } 3 \cdot \text{Row } 3 \\ = \begin{bmatrix} 1 & 2 \\ 3 & 6 \\ -1 & -2 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 6 \\ 0 & -6 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 2 & -1 \\ 2 & -1 \\ 2 & -1 \\ 10 & -5 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 5 & 11 \\ 1 & -9 \\ 10 & -5 \end{bmatrix}$$

10. All statements except the 1st one are equivalent to saying that A is invertible.